



Instructor: Date:

Learner: Regional Branch:

Program: Center:

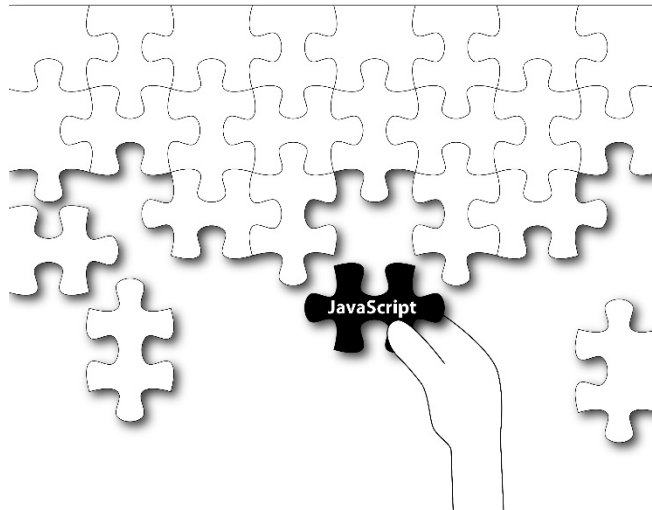
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Let's Start!

Let's talk about JavaScript!



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Explain briefly how different programming languages work.

Can Do Statements:

- I can explain what JavaScript is.
- I can describe how JavaScript works.
- I can identify and understand key features and concepts related to JavaScript.

Ask your partner:

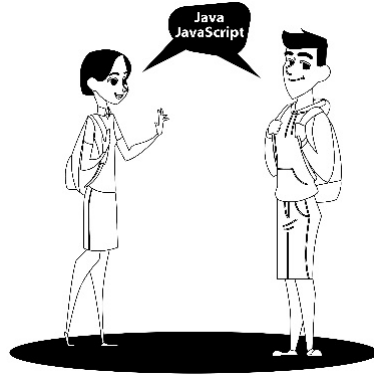
- What programming languages do you know?
- Have you worked with JavaScript before? How was it?





Let's Explore!

Christian is worried* because he doesn't remember much about JavaScript
**"worried"* is when you are concerned or anxious about something.



Instructions: Read the text below.

- Christian:** Hey, Danna! Is it true you know about JavaScript? Because I need help! I have to prepare a presentation about that topic and I'm a little bit rusty on that. Can you tell me what JavaScript is?
- Danna:** Sure, JavaScript is a type of general-purpose programming language standardized in the ECMAScript language specification. It is also one of the three core technologies on the World Wide Web, including HTML and CSS; we also consider it as generic software.
- Christian:** You should tell me about the key features of JavaScript as well, don't you think so?
- Danna:** Well, some of the key features of JavaScript are that it uses a GML, which is Generalized Markup Language. It is high-level; it is also dynamic and can work with regular expressions. Most web browsers support JavaScript so you don't have to download any plug-ins. Besides that, when you work with JavaScript, it lets you change the structure of the web at runtime, and not during the compilation.
- Christian:** Anything else I should know about JavaScript?
- Danna:** Well, yes. I have to tell you how it works.
- Christian:** Please do!
- Danna:** So, when you open up a web page, the browser has to fetch the embedded JavaScript; after that, and depending what you want the JavaScript to do, it will either execute as soon as the file loads or after a triggering event, like a click or certain amount of time. Finally, it executes the code and all of the changes you make will instantly take effect.
- Christian:** So, if I go to a website and load it, the browser has a built-in interpreter that reads the JavaScript code found on the page and runs it, right?
- Danna:** Correct!



Let's Practice!

Let's review some concepts related to java script

Instructions: Match column A with Column B.

COLUMN A

1. Specification
2. Regular expression
3. Generic software
4. Core technology

COLUMN B

- a. A standard that provides the necessary details about specific requirements.
- b. A sequence of characters used to define a search pattern.
- c. A technology that is essential for the functioning of the program.
- d. A software used for various purposes. It doesn't need to be modified.



Let's Practice!

Danna is helping Cristian to structure his presentation.

Instructions: Select the best options in the table. There is only one answer per sentence.

	a type of general-purpose programming language	technologies on the World Wide Web.	language specification, as every programming language.	is also considered to be a generic software.
1. First, you have to explain that Java is				
2. You should also say that it was standardize in the ECMAScript				
3. You must clarify that it is one of the three core				
4. You have to remind them that it				



Extension Activity

These are some relevant concepts to this context!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Choose 6 of the relevant concepts in computer science and Java Script.
- Summarize the information you found.
- Write a short paragraph using sentences that express advices and orders. For example: They must standardize every programming language.
- Share and compare your sentences with your partners.

Self-Assessment

- I understand relevant concepts in computer science and JavaScript.
- I summarize information related to this topic.
- I write sentences with orders and advices.
- I share and compare my sentences with my partners.

Yes No

☐ ☐
☐ ☐
☐ ☐
☐ ☐


Target Words

- **rusty** (adjective): when you are out of practice; impaired in skill or knowledge about something or a specific topic
e.g. I have to prepare a presentation about that topic and I'm a little bit rusty on that. Can you tell me what JavaScript is?
- **runtime** (noun): the time a computer program takes to be executed
e.g. It lets you change the structure of the web at runtime, and not during the compilation.
- **core technology** (noun): a technology that is essential for the functioning of the program
e.g. It is also one of the three core technologies on the World Wide Web, including HTML and CSS.
- **open up** (phrasal verb): to start something or to begin operation
e.g. So when you open up a web page the browser fetches the embedded JavaScript.
- **fetch** (verb): to go after something and return with it
e.g. So when you open up a web page the browser has to fetch the embedded JavaScript.
- **instantly** (adverb): when something is done at once or very fast
e.g. Finally, it executes the code and all of the changes you make will instantly take effect.



Learning Points

Modal Verbs for Obligation and Advice

Have to and **must** are modal verbs that express obligation. That means that it is mandatory to do something.

Should is also a modal verb, but it is commonly used to express advice, something that is a good idea.

When you want to say that something is mandatory or that is necessary to do something, you can use **have to** or **must**.

Follow this structure:

Subject + **have to/must** + **verb base form** + complement.

Example:

*I **have to** follow this guide to use JavaScript.*

*She **must** use JavaScript on that page.*

Note:

*When you use **have to** remember to change it to **has to** in the third person.*

Example:

*She **has to tell** us how to use JavaScript.*

Remember that when you use modal verbs, you don't need to add "-s" in the third person.

Example:

*Daniel **must** have the presentation ready for Monday.*

If you want to say that something is a good idea, or give some advice to somebody, you can use the modal **should**.

Example:

*They **should** teach about programming languages.*

Examples

- *You should tell me about the key features of JavaScript as well, don't you think so?*
- *Well, yes. I have to tell you how it works.*
- *So when you open up a web page the browser has to fetch the embedded JavaScript.*





Learning Points

Common Content Words and Expressions Used in JavaScript

When you use JavaScript, you may find some words that are common like: **High-level, dynamic, standardize, specification, core technology, regular expressions.**

In case you get to find these words when using JavaScript or maybe reading about it, here you have the meaning of each one of them.

High-level: When talking about computers, a high-level programming language is one that has strong abstraction from the details of the computer; it is easier to use and makes the process simpler.

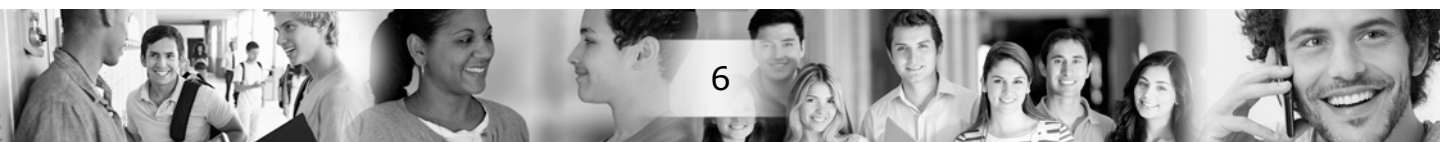
Dynamic: In computer science, a dynamic programming language is one that executes many programming behaviors at runtime, instead of doing it during compilation, as static programming languages do.

Standardize: To make everything into conformity with a standard.

Specification: Is a type of a standard which is often referenced by a contract or procurement document. It provides the necessary details about the specific requirements.

Core technology: A technology that is an important part of something else; it is essential for the functioning of the program.

Regular expression: It is a sequence of characters used to define a search pattern.





Learning Points

List of Programming and Computer Science Terms

In computer science, we can find several terms that are important to know, such as **Generic software**, **GML (Generalized Markup Language)** and **GPL (General Purpose Language)**.

Here are the definitions for each one of them.

Generic software: this is a kind of software that can be used for multiple purposes without having to modify it.

GML: (Generalized Markup Language) It is a set of macros that adds tags to a text and defines what it is in terms of paragraphs, headers, lists, tables, etc.

GPL: (General Purpose Language) This is when a programming language is designed for writing software in many application domains, not just one.

Examples

- *Well some of the key features of JavaScript are that it uses a GML, which is Generalized Markup Language.*
- *JavaScript is a type of general-purpose programming language that was standardized in the ECMAScript language specification.*